C^* Algebras and Gelfand Duality

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Everything I have written is my own work from my own understanding, however I have used a lot of sources, which I declare and explain in [1].

1 Introduction

C^* algebras are nice objects because they come with a lot of algebraic and analytic structure to work with. As a consequence, we have access to many theorems and properties derived from both Algebra and Analysis. This makes C^*-algebras particular useful in Physics, for example, when discussing bounded operators on some Hilbert space representing some Hamiltonian[9]. It turns out that commutative, unital C^* algebras are essentially equivalent to compact Hausdorff spaces and this essay builds up the theory behind and proves this theorem (known as Gelfand Duality). Further, the way in which they are equivalent is made precise in a categorical sense, and we end by establishing an equivalence of categories.

2 C* Algebras

From class, we are already familiar with the concept of normed algebras and Banach algebras. For instance, if X is a normed space, then $X^* = B(X, \mathbf{C})$ with the operator norm is an important example of a Banach algebra. Additionally, if X is a topological space, then we can construct $Bd(X, \mathbf{C})$, the space of totally bounded maps $f : X \to \mathbf{C}$. That is, maps such that $\sup_{x\in X} |f(x)| < \infty$. This is a Banach algebra with supremum norm $\|f\|_{\sup} := \sup_{x\in X} |f(x)|$ and multiplication defined point-wise. This example is important because we will constantly be referring to $C_0(X)$, the subalgebra of $Bd(X, \mathbf{C})$ of continuous functions which vanish at ∞.

Definition 2.1: A continuous function $f : X \to \mathbf{C}$ *vanishes at* ∞ if $|f|^{-1}([\varepsilon, \infty])$ is compact for all $\varepsilon > 0$.

Remark 2.2: If X is compact and Hausdorff, then $C_0(X) = C(X)$ because $|f|^{-1}([\varepsilon, \infty])$ will be a closed subset of X and hence always compact. To be most general, we should work with $C_0(X)$, especially for locally compact spaces X. However, we will mostly assume that any algebra A with work with is unital, which will cause the associated space with it X to be compact. So often we will drop the subscript zero in future examples.

Next we introduce the structure of an involution over A. This is a generalisation of the idea of adjoint maps in $B(H)$, the bounded linear maps on a Hilbert space.

Definition 2.3: Let A be an algebra. An *involution* over A is a map $* : A \to A$ such that for any $x, y \in A, \lambda \in \mathbf{C}$, the following holds

$$\textbf{i)}\ (x^*)^* = x, \qquad \textbf{ii)}\ (x+y)^* = x^* + y^*, \qquad \textbf{iii)}\ (\lambda x)^* = \overline{\lambda} x^*, \qquad \textbf{iv)}\ (xy)^* = y^* x^*$$

If A is a normed algebra, then we also require that $\|x^*\| = \|x\|$. We call A a $*-$algebra and we define a $*-$subalgebra of A to be a subalgebra of A that is closed under involution. Finally, we can now define the notion of a C^*-algebra.

Definition 2.4: An involutive Banach algebra is a C^**-algebra* if for any $x \in A$, $\|x^*x\| = \|x\|^2$. This equation is known as the C^**-identity*.

C^*-algebras are nice to work with because they have a lot of structure, and the C^*-identity gives us well behaved properties of the norm. For instance, if A is unital then the C^*-identity enforces that $\|1\| = 1$ (where $1 \in A$ is the unit of A)[1]. A simple example of a C^*-algebra that we are very familiar with is the bounded linear operators $B(H)$ on some Hilbert space. Here, the involution is taking a map $T \in B(H)$ to its adjoint $T^* \in B(H)$ and this satisfies the criterion that $\|TT^*\| = \|T\|^2$ for any $T \in B(H)$. Further, it is (somewhat surprisingly) true that any C^*-algebra is $*$-isomorphic to a closed sub-algebra of $B(H)$ for a suitable Hilbert space H. This is the statement of the Gelfand-Naimark theorem[3], which is to some degree related to Gelfand duality.

Another example of a C^*-algebra is $C_0(X)$ for any locally compact, Hausdorff space X under the supremum norm[1]. If X is compact, then $C(X)$ is unital because the map $f(x) = 1$ is an element of $C(X)$. Whereas, if X is not compact, then f may not vanish at infinity and so $C_0(X)$ may not be unital. To avoid any difficulties that come with not having a unit and to simplify a lot of proofs, from now on we are going to assume that any algebra we talk about is unital. This does not lose much generality because almost all the following proofs can be generalised to the non-unital case without too much difficulty by passing to the unitisation[1]. The unitisation A_1 of an algebra A is an enlarged version of A that adds in a unit and has A isometrically isomorphic to a sub-algebra[2] of A_1.

3 The Spectrum of a Banach Algebra

Since we are now working with unital algebras A, we can talk about inverses of elements of A. Denote the invertible elements of A as $A^\times$. This is a group under multiplication and it is an open subset of A.

Proposition 3.1: Let A be a unital Banach algebra. If $\|x - 1\| < 1$, then $x \in A^\times$. Further, if $a \in A^\times$ and $\|x - a\| < \frac{1}{\|a^{-1}\|}$, then $x \in A^\times$ and $x^{-1} = \sum_{n=0}^{\infty} a^{-1}(1 - xa^{-1})^n$.

Proof. Suppose $\|x - 1\| < 1$ and let $m \in \mathbf{N}$. Then notice that we have

$$\begin{aligned} x\sum_{n=0}^{m}(1-x)^n &= (1-(1-x))\sum_{n=0}^{m}(1-x)^n \\ &= 1-(1-x)^{m+1} \\ &\to 1 \end{aligned}$$

Since $\|x-1\| < 1$, we have that $(1-x)^{m+1} \to 0$ as $m \to \infty$. This shows that $x^{-1} = \sum_{n=0}^{\infty}(1-x)^n$. Now suppose that $\|x - a\| < \frac{1}{\|a^{-1}\|}$ for some $a \in A^\times$. Then $\|xa^{-1} - 1\| = \|(x - a)a^{-1}\| \leq \|x - a\|\|a^{-1}\| < 1$. Applying the above argument to xa^{-1}, we find that $(xa^{-1})^{-1} = \sum_{n=0}^{\infty}(1 - xa^{-1})^n$ and so $x^{-1} = \sum_{n=0}^{\infty} a^{-1}(1 - xa^{-1})^n$. $\square$

This proposition shows that $A^\times$ is open since any element $a \in A$ has a ball of radius $\frac{1}{\|a^{-1}\|}$ contained in A. Understanding the invertible elements of A is important in understanding the spectrum of elements of A.

Definition 3.2: Let A be a unital algebra and $a \in A$. The *spectrum* of $a \in A$ is

$$\sigma_A(a) := \{\lambda \in \mathbf{C} \mid a - \lambda \notin A^\times\}$$

Additionally, we will call the complement of $\sigma_A(a)$ in $\mathbf{C}$ the resolvent of a in A, and denote this as $\mathrm{Res}_A(a)$.

To simplify notation, we will often drop the subscript A when it is clear what algebra we are talking about. Also, notice that in the above definition, we wrote λ, when we really meant $\lambda \cdot 1$, the inclusion of λ into A. If we let our algebra A be $M_n(\mathbf{C})$, the $n \times n$ matrices with coefficients in $\mathbf{C}$, then the spectrum of a matrix $M \in A$ is exactly the set of eigenvalues of M. The next few propositions and definitions are designed for us to build up enough background to prove the Gelfand-Mazur Theorem.

Theorem 3.3 (Gelfand-Mazur)**:** If a Banach algebra A is a division ring, the it is isomorphic to $\mathbf{C}$.

To do this, we define the spectral radius of an element of a Banach algebra and show that it has some important properties.

Definition 3.4: The *spectral radius* of an element a of a Banach algebra A is $r_A(a) := \sup\{|\lambda| \mid \lambda \in \sigma_A(a)\}$

In the finite dimensional case of $A = M_n(\mathbf{C})$ as mentioned above, $r_A(M)$ will simply be the eigenvalue of M with largest norm, since $\sigma_A(M)$ is a finite set. Notice that the set $\sigma_A(a)$ may be empty and in this case, we would have $r_A(a) = -\infty$. We want $r_A(a) \geq 0$ for all $a \in A$, and so later we show that $\sigma_A(a)$ is never empty.

Proposition 3.5: Let A be a Banach algebra and $a \in A$. Then $r(a) \leq \|a\|$.

Proof. Suppose that $\lambda \in \mathbf{C}$ such that $|\lambda| > \|a\|$. Then $\|1 - (1 - \frac{a}{\lambda})\| = \|\frac{a}{\lambda}\| < 1$. Therefore, by Proposition 3.1, $1 - \frac{a}{\lambda} = -\frac{a-\lambda}{\lambda}$ is invertible which means that $a - \lambda$ is invertible. That is, $\lambda \in \text{Res}(a) = \sigma(a)^C$. So for all $\lambda \in \sigma(a)$, we have $|\lambda| \leq \|a\|$ and by taking the supremum, we get $r(a) \leq \|a\|$. □

Proposition 3.6: Let A be an algebra and $a \in A$. If $\lambda \in \sigma(a)$, then $\lambda^n \in \sigma(a^n)$ for all $n \in \mathbf{N}$.

Proof. Suppose $\lambda \in \sigma(a)$. Notice that we can write $a^n - \lambda^n$ as

$$a^n - \lambda^n = (a - \lambda) \sum_{i=0}^{n-1} a^{(n-1)-i} \lambda^i$$

From this equation, if $a^n - \lambda^n$ is invertible, then so is $a - \lambda$. Therefore $a^n - \lambda^n$ is not invertible, so $\lambda^n \in \sigma(a^n)$.□

Proposition 3.7: Let A be a Banach algebra. For all $a \in A$, $\|a^n\|^{\frac{1}{n}} \to r(a)$ as $n \to \infty$.

Proof. To do this, we are going to show that $\limsup_{n\to\infty} \|a^n\|^{\frac{1}{n}} \leq r(a) \leq \liminf_{n\to\infty} \|a^n\|^{\frac{1}{n}}$. Fix some $a \in A$ and let $\lambda \in \sigma(a)$. Then $\lambda^n \in \sigma(a^n)$ by Proposition 3.6 and then applying Proposition 3.5, we have $|\lambda|^n \leq r(a^n) \leq \|a^n\|$ and therefore $r(a) \leq \|a^n\|^{\frac{1}{n}}$. That is, we have $r(a) \leq \liminf_{n\to\infty} \|a^n\|^{\frac{1}{n}}$ which gives us the RHS inequality.

To obtain the other inequality, we are going to show that $\sum_{n=0}^{\infty} \frac{a^n}{\lambda^{n+1}}$ is absolutely convergent for all $\lambda \in \mathbf{C}$ with $|\lambda| > r(a)$. This proves the LHS inequality because, by the n-th root test, we would have $\limsup_{n\to\infty} (\frac{\|a^n\|}{\lambda^{n+1}})^{\frac{1}{n}} \leq 1$ and therefore $\limsup_{n\to\infty} \|a^n\|^{\frac{1}{n}} \leq |\lambda| \leq r(a)$.

To prove that $\sum_{n=0}^{\infty} \frac{a^n}{\lambda^{n+1}}$ is absolutely convergent for all $|\lambda| > r(a)$, we will need to use some complex analysis[4]. Define a function $f : \mathbf{C} \to A$ by $f(\mu) = \frac{1}{a-\mu}$. This function is well defined on $\text{Res}(a)$ since $a - \mu$ is invertible for any $\mu \in \text{Res}(a)$. We are going to show f is holomorphic on $\text{Res}(a)$. First notice that if $|\mu - \mu_0| < \|a - \mu_0\|$ for some fixed $\mu_0 \in \text{Res}(a)$, then $\|(a - \mu_0) - (a - \mu)\| < \|a - \mu_0\|$ and so $\|1 - \frac{a-\mu}{a-\mu_0}\| < 1$. By proposition 3.1, this ensures that $\frac{a-\mu}{a-\mu_0}$ is invertible and hence $a - \mu$ is invertible. Further, we can calculate its inverse as

$$\begin{aligned}
(a-\mu)^{-1} &= \left(\frac{a-\mu}{a-\mu_0}\right)^{-1} \cdot (a-\mu_0)^{-1} \\
&= \sum_{n=0}^{\infty} \left(1 - \frac{a-\mu}{a-\mu_0}\right)^n \cdot \frac{1}{a-\mu_0} \\
&= \sum_{n=0}^{\infty} (\mu-\mu_0)^n f(\mu_0)^{n+1}.
\end{aligned}$$

This series is absolutely convergent so long as $|\mu - \mu_0| < \|a - \mu_0\|$ which means that f is holomorphic on an open neighbourhood of μ. Therefore, f is holomorphic on $\text{Res}(a)$. Now let $\lambda \in \mathbf{C}$ such that $|\lambda| > \|a\|$. This means that $\sum_{n=0}^{\infty} \frac{-a^n}{\lambda^{n+1}}$ is absolutely convergent because it is the sum of a geometric series, and it converges to $-\frac{1}{\lambda}(1 - \frac{a}{\lambda})^{-1} = (a - \lambda)^{-1}$. This means that $\sum_{n=0}^{\infty} \frac{-a^n}{\lambda^{n+1}}$ is the Laurentz expansion of f about ∞. Since f is holomorphic on $\text{Res}(a)$, we have that it is holomorphic on $\{\lambda \mid |\lambda| > r(a)\}$ which means that $\sum_{n=0}^{\infty} \frac{-a^n}{\lambda^{n+1}}$ converges for $|\lambda| > r(a)$, which is what we were required to show. □

Proposition 3.8: Let A be a Banach algebra. Then $\sigma(a)$ is non-empty for all $a \in A$.

Proof. Let $a \in A$ and let f be the function defined in the previous proposition. Let $\varphi \in A^*$ and define $f_\varphi = \varphi \circ f : \text{Res}(a) \to \mathbf{C}$. From before, we know that f_φ is holomorphic on $\text{Res}(a)$. Suppose, in order to get a contradiction, that $\sigma(a)$ is empty. Then $\text{Res}(a) = \mathbf{C}$ and therefore f is entire. Additionally, we can see that $\lim_{|\lambda|\to\infty} f_\varphi(\lambda) = \lim_{|\lambda|\to\infty} \varphi(\frac{1}{a-\lambda}) = 0$. Therefore, f_φ is a bounded entire function and hence by the Liouville theorem[4], f_φ is a constant function. Further, since $f_\varphi(\lambda) \to 0$, f_φ must be the zero function. Since this is true for all $\varphi \in A^*$, we have that f is the zero function. But this a contradiction since $f(\lambda) = (a - \lambda)^{-1}$ and $(a - \lambda)^{-1}$ cannot be equal to zero because it is invertible. Therefore $\sigma(a)$ is non-empty. □

Corollary 3.9 (Gelfand Mazur)**:** If a Banach algebra A is a division ring, the it is isomorphic to $\mathbf{C}$.

Proof. We already know that $\mathbf{C}$ includes into A by the map $i : \mathbf{C} \to A$, $i(\lambda) = \lambda \cdot 1$ and this is a continuous norm 1 injection. Suppose A is a division ring and let $a \in A$. By Proposition 3.8 there exists $\lambda \in \sigma(a)$ such that $a - \lambda$ is not invertible. Since A is a division ring, the only non-invertible element in A is zero so $a - \lambda = 0$. That is, $a \in i(\mathbf{C})$ and therefore $A = i(\mathbf{C})$. Since $i(\mathbf{C}) \cong \mathbf{C}$, we have $A \cong \mathbf{C}$. □

4 Characters and Maximal Ideals

To construct the Gelfand Transfrom, we need to look at the characters of an algebra A, which are the non-zero homomorphisms from A into $\mathbf{C}$. It turns out that these characters are in bijective correspondence with maximal ideals of A and this correspondence relies on the Gelfand Mazur theorem that we have just proven. We will spend the rest of this section constructing this correspondence, and then use it to define the Gelfand transform. Note here that whenever we say ideal of A, we mean a two sided ideal, which is important because we are not currently assuming that our algebras are commutative.

Definition 4.1: An ideal $I \subset A$ is *proper* if $I \neq A$. A proper ideal m is *maximal* if $m \not\subseteq I$ for any proper ideal I.

Lemma 4.2: Any maximal ideal $m \subset A$ is closed.

Proof. A maximal ideal m cannot be dense in A because $A^\times$ is open in A (Proposition 3.1) and m does not intersect $A^\times$ (otherwise m is not proper). Further, $m \subset \overline{m}$ and $\overline{m}$ is an ideal of A (because limits are linear). If m is not dense, $\overline{m} \neq A$, hence $\overline{m} = m$ by the maximality of m and so m is closed. □

Definition 4.3: Let A and B be Banach algebras. A continuous linear map $\varphi : A \to B$ is a *homomorphism* if $\varphi(aa') = \varphi(a)\varphi(a')$ for all $a, a' \in A$. A homomorphism which is not necessarily continuous is called an *algebraic homomorphism.* If A and B are involutive algebras, then we also require that $\varphi(a^*) = \varphi(a)^*$. In this case φ is a **-homomorphism.*

Proposition 4.4: Let I be a closed ideal of A. Then A/I is a Banach algebra under the standard quotient norm and $\pi : A \to A/I$ is a homomorphism with $\|\pi\| \leq 1$.

Proof. From class we already know that A/I is a Banach space under the standard quotient norm and that π is continuous with $\|\pi\| \leq 1$. The definition of π clearly shows that π is a homomorphism. To show that A/I is an algebra, we need to show that the multiplication structure induced by π is sub-multiplicative. Let $\varepsilon > 0$ and $\pi(a), \pi(a')$ be elements of A/I with representatives $a, a' \in A$. Choose $i, j \in I$ such that $\|a + i\| \leq \|\pi(a)\| + \varepsilon$ and $\|a' + j\| \leq \|\pi(a')\| + \varepsilon$ (this is possible by the infimum definition of the norm on A/I).

$$\begin{aligned}\|\pi(a)\pi(a')\| &= \|\pi((a + i)(a' + j))\| \\ &\leq \|\pi\|\|a + i\|\|a + j\| \\ &\leq \|\pi(a)\|\|\pi(a')\| + \varepsilon(\|\pi(a)\| + \|\pi(a')\| + \varepsilon)\end{aligned}$$

Letting $\varepsilon \to 0$ we have shown that $\|\pi(a)\pi(a')\| \leq \|\pi(a)\|\|\pi(a')\|$. □

Proposition 4.5: Every proper ideal I of a commutative algebra A is contained in a maximal ideal of A.

Proof. Consider the set of proper ideals of A that contain I, partially ordered by inclusion. Every chain in this ordering has an upper bound, namely the union of all the ideals in the chain (which is still a proper ideal). Therefore, by Zorn's lemma, there exists a maximal element m in this set. That is, a proper ideal m such that m contains I and m is not contained in any other proper ideal. So m is a maximal ideal containing I. □

Corollary 4.6: Every non-invertible element of a commutative algebra A is contained in a maximal ideal.

Proof. If $a \in A$ is not-invertible, the ideal generated by a is proper. Use Proposition 4.5. □

Now we have enough background to construct the correspondence between $M(A) := \{\text{maximal ideals of } A\}$ and the characters of A, $\Omega(A) := \{\omega : A \to \mathbf{C}, \omega \neq 0\}$. If A is a commutative Banach algebra, then an ideal $m \subset A$

is maximal if and only if A/m is a field. By Proposition 4.4, A/m is also a Banach algebra and therefore by the Gelfand-Mazur Theorem (3.3), $A/m \cong \mathbf{C}$. Thus, there is a $\mathbf{C}$-algebra isomorphism $\tilde{\omega}_m : A/m \to \mathbf{C}$ and it is unique because it is determined by where $\overline{1} \in A/m$ is sent, but $\overline{1}$ must be sent to $1 \in \mathbf{C}$. Composing with $\pi : A \to A/m$, we have a unique map $\omega_m := \tilde{\omega}_m \circ \pi : A \to \mathbf{C}$ that factors through A/m. That is, for any $m \in M(A)$, we can construct a unique $\omega_m \in \Omega(A)$ that depends solely on m. Similarly, if $\omega \in \Omega(A)$, then ω is surjective since it is not zero.

$$\begin{tikzcd}
A \arrow[rr, two heads, "\omega"] \arrow[dr, "\pi"'] & & \mathbf{C} \\
& A/\ker\omega \arrow[ur, dashed, "\sim", "\exists!\tilde{\omega}"'] &
\end{tikzcd}$$

By the first isomorphism theorem, $A/\ker\omega \cong \mathbf{C}$ and therefore $m_\omega := \ker\omega$ is a maximal ideal. This is the reverse of our correspondence. That is, the maps $F : M(A) \to \Omega(A)$, $F(m) = \omega_m$ and $G : \Omega(A) \to M(A)$, $G(\omega) = m_\omega$ are inverses of each other.

Proposition 4.7: Let A be a commutative Banach algebra. The the map $F : M(A) \to \Omega(A)$ mentioned above is a bijection of sets with inverse $G : \Omega \to M(A)$ also defined above.

Proof. Let $m \in M(A)$ and $F(m) = \omega_m : A \to \mathbf{C}$. We have that $G(\omega_m) = \ker\omega_m$. But $\omega_m = \tilde{\omega}_m \circ \pi$ and $\tilde{\omega}_m$ is an isomorphism. Therefore $\ker\omega_m = \ker\pi = m$ so $GF = \mathrm{id}_{M(A)}$. Now let $\omega \in \Omega(A)$. Then $G(\omega) = m_\omega = \ker\omega$. We construct $F(\ker\omega)$ by composing π with the unique isomorphism $\tilde{\omega}_{\ker\omega} : A/\ker\omega \to \mathbf{C}$. But we also get an isomorphism $\varphi : A/\ker\omega \to \mathbf{C}$ from the first isomorphism theorem and $\omega = \varphi \circ \pi$. Therefore by the uniqueness of $\tilde{\omega}_{\ker\omega}$, $F(\ker\omega) = \tilde{\omega}_{\ker\omega} \circ \pi = \varphi \circ \pi = \omega$. That is, $FG = \mathrm{id}_{\Omega(A)}$. □

5 The Gelfand Transform

Our overall goal is to show that C^*-algebras are equivalent in some sense to compact Hausdorff topological spaces. So now we show that $\Omega(A)$ is compact and Hausdorff in the weak-* topology on A^*.

Proposition 5.1: Let A be a commutative Banach algebra. Then $\Omega(A) \subset (A^*)_1$, the norm-closed unit ball of A^*, and $\Omega(A)$ is compact and Hausdorff in the weak-* topology of A^*.

Proof. Let $\omega \in \Omega(A)$. For any $a \in A$, we have $|\omega(a)| = |\omega(a^n)|^{\frac{1}{n}} \leq \|\omega\|^{\frac{1}{n}}\|a^n\|^{\frac{1}{n}}$. Therefore, $|\omega(a)| \leq \lim_{n\to\infty} \|\omega\|^{\frac{1}{n}}\|a^n\|^{\frac{1}{n}} = r(a)$ by Proposition 3.7. By Proposition 3.5, $r(a) \leq \|a\|$ and therefore $\|\omega\| \leq 1$. So $\omega \in (A^*)_1$. Now let $\Omega^1(A) = \Omega(A) \cup \{0\}$. We will show that $\Omega^1(A)$ is closed in the weak-* topology by showing that the limit of any weakly convergent net[5] in $\Omega^1(A)$ is contained in $\Omega^1(A)$. So let $\{\omega_i\}$ be a net in $\Omega^1(A)$ converging to $\omega_0 \in (A^*)_1$.

$$\begin{aligned}
\omega_0(ab) &= \lim_i \omega_i(ab) \\
&= \lim_i \omega_i(a)\omega_i(b) \\
&= \omega_0(a)\omega(b)
\end{aligned}$$

This last equality holds because weak-* convergence is point-wise convergence. This shows that ω_0 is a homomorphism and therefore $\omega_0 \in \Omega^1(A)$. That is, $\Omega^1(A)$ is a weak-* closed subset of $(A^*)_1$ and therefore is compact and Hausdorff in the weak-* topology. For any $\omega \in \Omega(A)$, we have $\omega(1) = 1$. Therefore, $\|\omega - 0\| \geq 1$ so zero is an isolated point of $\Omega^1(A)$. Thus, $\Omega(A)$ is a weak-* closed subset of $\Omega^1(A)$ and $\Omega(A)$ is weak-* compact and Hausdorff. □

Definition 5.2 (The Gelfand Transform)**:** Let A be a commutative Banach algebra. For any $a \in A$, define $\hat{a} \in C(\Omega(A))$ by $\hat{a}(\omega) := \omega(a)$ for all $\omega \in \Omega(A)$. The map $\mathcal{G} : A \to C(\Omega(A))$, $\mathcal{G}(a) = \hat{a}$ is called the *Gelfand transform*.

It should be noted that each $\hat{a}$ is continuous in the weak-* topology because the weak-* topology is defined to make the evaluation maps continuous. This means that $\mathcal{G}$ is a well-defined map.

Theorem 5.3: Let A be a commutative Banach algebra and $a \in A$.

1. $\sigma(a) = \hat{a}(\Omega(A))$.
2. $r(a) = \|\hat{a}\|_{\sup}$
3. The Gelfand transform is a continuous homomorphism and $\|\hat{a}\|_{\sup} \leq \|a\|$.

Proof. 1. Let $\lambda \in \sigma(a)$. That is, $a - \lambda$ is not invertible. Corollary 4.6 implies that $a - \lambda \in m$ for some maximal ideal $m \subset A$. Therefore $\omega_m(a - \lambda) = 0$, where ω_m is defined in Proposition 4.7. Thus, $\omega_m(a) = \lambda\omega_m(1) = \lambda$ and so $\hat{a}(\omega_m) = \lambda$. That is, $\lambda \in \hat{a}(\Omega(A))$ and $\sigma(a) \subset \hat{a}(\Omega(A))$. Now suppose that $\lambda \in \hat{a}(\Omega(A))$ with $\lambda = \hat{a}(\omega)$ for some $\omega \in \Omega(A)$. Then $\omega(a - \lambda) = 0$ and so $a - \lambda \in m_\omega$ where m_ω is the maximal ideal defined in Proposition 4.7. Since m_ω is maximal, $a - \lambda$ cannot be invertible and therefore $\lambda \in \sigma(a)$. That is, $\hat{a}(\Omega(A)) \subset \sigma(a)$ and thus $\sigma(a) = \hat{a}(\Omega(A))$.

2. Let $\hat{a} \in C(\Omega(A))$.

$$\begin{aligned}\|\hat{a}\|_{\sup} &= \sup\{|\hat{a}(\omega)| \mid \omega \in \sigma(a)\}\\ &= \sup\{|\lambda| \mid \lambda \in \hat{a}(\Omega(A)) = \sigma(a)\}\\ &= r(a)\end{aligned}$$

3. It is clear that $\mathcal{G}$ is linear, sub-multiplicative and an algebraic homomorphism from the definition of evaluation maps. From above, we have that $\|\hat{a}\|_{\sup} = r(a) \leq \|a\|$ by Proposition 3.5 which shows that $\mathcal{G}$ is bounded, hence continuous. So $\mathcal{G}$ is a homomorphism. □

6 Gelfand Duality

With the Gelfand transform defined we are now ready to prove that commutative C^*-algebras are isometrically *-isomorphic to continuous functions on a compact space.

Definition 6.1: A commutative involutive Banach algebra A is *symmetric* if $\omega(a^*) = \overline{\omega(a)}$ for all $\omega \in \Omega(A)$ and $a \in A$.

Definition 6.2: Let A be an involutive Banach algebra.

- $a \in A$ is *self-adjoint* if $a = a^*$.
- $a \in A$ is *normal* if $aa^* = a^*a$.
- $a \in A$ is *unitary* if $aa^* = a^*a = 1$.

These two definitions are very intuitive and familiar if we think about the $n \times n$ matrices case. If $A = M_n(\mathbf{C})$, then the involution operation is taking the conjugate transpose, and the notions of self-adjoint, normal and unitary all agree with our previously understood notions. These definitions just generalise them.

Proposition 6.3: If A is a symmetric Banach algebra, then $\hat{A}$ is dense in $C(\Omega(A))$.

Proof. This proof relies on the Stone-Weirstrauss theorem[6] which states if X is compact and Hausdorff and B is a *-subalgebra of $C(X)$ which separates points and vanishes nowhere, then B is dense in $C(X)$. Since we know that $\Omega(A)$ is compact, it suffices to show that $\hat{A}$ separates points and vanishes nowhere. We have $\hat{A}$ separates points because if $\omega_1 \neq \omega_2$, then there exists $a \in A$ such that $\omega_1(a) \neq \omega_2(a)$ and so $\hat{a}(\omega_1) \neq \hat{a}(\omega_2)$. Additionally, $\hat{A}$ vanishes nowhere since for all $\omega \in \Omega(A)$, $\omega \neq 0$ so there exists $a \in A$ such that $\omega(a) \neq 0$. That is, $\hat{a}(\omega) \neq 0$ so $\hat{a} \neq 0$. Since A is symmetric, $\hat{A}$ is a *-subalgebra of $C(\Omega(A))$ and therefore the result follows from the Stone-Weirstrauss theorem. □

This proposition is going to be very useful later, but in order to ever hope of using it, we first need to show that commutative C^* algebras are symmetric.

Lemma 6.4: Let A be a commutative, involution Banach algebra. Then the following are equivalent.

1. A is symmetric.
2. $\widehat{a^*} = \overline{\hat{a}}$ for all $a \in A$.
3. $\omega(a) \in \mathbf{R}$ for all $\omega \in \Omega(A)$ and self-adjoint $a \in A$.

Proof. ($1 \Rightarrow 2$) is clear since if A is symmetric, $\widehat{a^*}(\omega) = \omega(a^*) = \overline{\omega(a)} = \overline{\hat{a}}(\omega)$. The implication ($2 \Rightarrow 3$) is also clear since if a is self-adjoint, $\omega(a) = \omega(a^*) = \widehat{a^*}(\omega) = \overline{\hat{a}}(\omega) = \overline{\omega(a)}$. To see ($3 \Rightarrow 1$), define $a_1 = \frac{a+a^*}{2}$ and $a_2 = \frac{a-a^*}{2i}$. Notice that a_1 and a_2 are self-adjoint and $a = a_1 + ia_2$, $a^* = a_1 - ia_2$. Thus $\overline{\omega(a)} = \overline{\omega(a_1) + i\omega(a_2)} = \omega(a_1) - i\omega(a_2) = \omega(a^*)$. □

Proposition 6.5: Every commutative C^*-algebra A is symmetric.

Proof. Let $a \in A$ be self-adjoint and $\omega(a) = x + iy \in \mathbf{C}$. Define $a_t = a + it$ with $t \in \mathbf{R}$. We have that $\omega(a_t) = \omega(a) + it = x + i(y+t)$ so notice that $|\omega(a_t)|^2 = x^2 + (y+t)^2 = \|aa^*\| + 2yt + t^2$. Further, we also have that $|\omega(a_t)|^2 \leq \|a_t\|^2\|1\| = \|a_t a_t^*\| = \|a^2 + t^2\| \leq \|a^2\| + t^2$. Therefore, $\|a\|^2 + 2yt \leq \|a\|^2$ for all $t \in \mathbf{R}$ which is only possible if $y = 0$. So $\omega(a)$ is real and thus by Lemma 6.4, we have that A is symmetric. □

The last piece we need is the following proposition and this will be used to show that the Gelfand transform is an isometery.

Proposition 6.6: Let A be a C^*-algebra and $a \in A$ a normal element. Then $r(a) = \|a\|$.

Proof. First we claim that $\|a^{2^n}\| = \|a\|^{2^n}$ for any $n \geq 0$ and some fixed normal element $a \in A$. Notice that this claim is true for any self-adjoint $b \in A$ because $\|b^2\| = \|bb^*\| = \|b\|^2$ and this can be easily extended by induction.

$$\begin{aligned}
\|a^{2^n}\|^2 &= \|(a^*)^{2^n} a^{2^n}\| \\
&= \|(a^*a)^{2^n}\| && \text{(because } a \text{ is normal)} \\
&= \|a^*a\|^{2^n} && (a^*a \text{ is self-adjoint}) \\
&= \left(\|a\|^{2^n}\right)^2 \\
\therefore \|a^{2^n}\| &= \|a\|^{2^n}
\end{aligned}$$

We know from Proposition 3.7 that $r(a) = \lim_{n\to\infty} \|a^n\|^{\frac{1}{n}}$ and therefore $r(a) = \lim_{n\to\infty} \|a^{2^n}\|^{\frac{1}{2^n}} = \lim_{n\to\infty} \|a\|^{\frac{2^n}{2^n}} = \|a\|$. □

Finally, we are now ready to put everything together and prove the statement of Gelfand Duality.

Theorem 6.7 (Gelfand Duality)**:** Let A be a commutative C^*-algebra. Then the Gelfand transform is an isometric *-isomorphism from A onto $C(\Omega(A))$.

Proof. Since A is commutative, all elements $a \in A$ are normal and therefore $\|a\| = r(a)$ by Proposition 6.6. By Theorem 5.3, $r(a) = \|\hat{a}\|_{\sup}$ and therefore $\|a\| = \|\hat{a}\|_{\sup}$. That is, $\mathcal{G}$ is an isometry. Since A is symmetric, we have by Lemma 6.4 that $\widehat{a^*} = \overline{\hat{a}}$, so $\mathcal{G}$ is a *-homomorphism. Since $\mathcal{G}$ is a continuous isometry, its image is closed. But, its image $\hat{A}$ is also dense in $C(\Omega(A))$ by Proposition 6.3. Therefore im $\mathcal{G} = C(\Omega(A))$ and $\mathcal{G}$ is surjective. We have already shown that $\mathcal{G}$ is an isometry, so it is injective. Therefore, $\mathcal{G}$ is a continuous linear isomorphism onto $C(\Omega(A))$ and by the Bounded Inverse Theorem, the inverse of $\mathcal{G}$ is bounded. Thus, $\mathcal{G}$ is an isometric *-isomorphism. □

7 An Equivalence of Categories

We have shown that any C^*-algebra is isometrically isomorphic to $C(\Omega(A))$, a compact Hausdorff topological space, but this correspondence also goes the other way round. Any compact Hausdorff topological space X is homeomorphic to $\Omega(C(X))$.

Theorem 7.1 (The Inverse Gelfand Transform)**:** Let X be a compact, Hausdorff topological space. Then the map $\mathcal{F} : X \to \Omega(C(X))$ defined by $\mathcal{F}(x) = \hat{x}$ where $\hat{x}(f) := f(x)$, $f \in C(X)$, is a homeomorphism.

Proof. First we show that $\mathcal{F}$ is well defined and $\hat{x} \in \Omega(C(X))$. It is multiplicative since $\hat{x}(f \cdot g) = (f \cdot g)(x) = f(x) \cdot g(x) = \hat{x}(f) \cdot \hat{x}(g)$. Similarly, $\hat{x}$ is also linear. This map is bounded since $|\hat{x}(f)| = |f(x)| \leq \|f\|_{\text{sup}}$. Since X is compact and Hausdorff, for any $x \in X$ we can use Urysohn's Lemma[7] to find a continuous function $f \in C(X)$ such that $f(x) \neq 0$. So $\hat{x}$ is never zero and thus $\mathcal{F}$ is well defined.

To see that $\mathcal{F}$ is injective, suppose $\hat{x}_1 = \hat{x}_2$. Then $f(x_1) = f(x_2)$ and again by Uryshon's Lemma[7], we have that $x_1 = x_2$. To see that $\mathcal{F}$ is surjective, let $\omega \in \Omega(C(X))$ and note that $\Omega(C(X)) = C(X)^* \setminus \{0\}$. Therefore, by the Riesz representation theorem, there exists a unique Boreal measure μ on X such that $\omega(f) = \int_X f d\mu$ for any $f \in C(X)$. Therefore $\omega(f - \omega(f)) = \int_X f - \omega(f) d\mu = 0$ and so

$$\begin{aligned}\int_X |f(x) - \omega(f)|^2 d\mu &= \omega\left((\overline{f - \omega(f)})(f - \omega(f)\right) \\ &= \omega(\overline{f - \omega(f)}) \cdot \omega(f - \omega(f)) \\ &= 0.\end{aligned}$$

That is, f is equal to the constant $\omega(f)$ $\mu-$almost everywhere. This means that μ is concentrated[1] at some $x_0 \in X$. Therefore $\int_X f d\mu = f(x_0)$ and so $\omega(f) = \hat{x}_0(f)$ for all $f \in C(X)$. That is, $\mathcal{F}$ is surjective.

To show that $\mathcal{F}$ is continuous, let x_i be a net[5] in X converging to x_0. Let $f \in C(X)$. Since f is continuous, we have $f(x_i) \to f(x_0)$. This holds for all $f \in C(X)$ and hence $\hat{x}_i \to \hat{x}_0$ in the weak-* topology. So $\mathcal{F}$ is continuous. Since $\mathcal{F}$ is a continuous bijection from a compact space to a Hausdorff space, we have that $\mathcal{F}$ is a homeomorphism[8]. □

We have already established that given two compact, Haussdorff spaces X and Y, we can get two commutative C^*-algebras $C(X)$ and $C(Y)$. However, if we are given a continuous map $\psi : X \to Y$, then we also have an induced *-homomorphism $\psi^* : C(Y) \to C(X)$ defined by $\psi^*(f) = f \circ \psi$. This induced map has nice properties, which we will demonstrate below.

Proposition 7.2: Let $\psi : X \to Y$ be a continuous map between compact Hausdorff topological spaces. The map $\psi^* : C(Y) \to C(X)$ induced by pre-composition is a $*$-homomorphism that respects composition and $\text{id}_X^* = \text{id}_{C(X)}$. Further, if ψ is a homeomorphism, then ψ^* is an isometric isomorphism.

Proof. If $f \in C(Y)$ is continuous, then certainly $f \circ \psi$ is continuous. Further, $\psi^*(f \cdot g) = (f \cdot g) \circ \psi = (f \circ \psi) \cdot (g \circ \psi) = \psi(f) \cdot \psi(g)$, so ψ^* is multiplicative and similarly linear. That is, ψ is a homomorphism. It is a *-homomorphism since $[\psi^*(\overline{f})](y) = \overline{f(\psi(y))} = \overline{\psi^*(f)}$. So this induced map is well defined.

Notice that $\text{id}|_X^*(f) = f \circ \text{id}|_X = \text{id}_X$ so taking this induced map preserves identities. Further if $\psi : X \to Y$ and $\phi : Y \to Z$, then $(\phi \circ \psi)^*(f) = f \circ \phi \circ \psi = \phi^*(f) \circ \psi = (\psi^* \circ \phi^*)(f)$ (notice that the order has been reversed). If ψ is onto, then $\|\psi^*(f)\| = \sup_{x \in X} |(f \circ \psi)(x)| = \sup_{x \in X} |f(x)| = \|f\|_{\text{sup}}$, so ψ^* is norm preserving. If ψ is a homeomorphism, then ψ^* is an isomorphism with $(\psi^*)^{-1} = (\psi^{-1})^*$. This is because $(\psi^{-1})^* \circ \psi^* = (\psi \circ \psi^{-1})^* = \text{id}^* = \text{id}$ and similarly $\psi^* \circ (\psi^{-1})^* = \text{id}$. Thus if ψ is a homeomorphism, ψ^* is an isometric isomorphism. □

In the exact same fashion, if $\psi : A \to B$ is a $*-$homomorphism between C^*-algebras, then we have an induced map $\psi^* : \Omega(B) \to \Omega(A)$ defined by $\psi^*(\omega) = \omega \circ \psi$. For the exact same reasons as above, ψ^* is well defined, respects composition and identities, and is a homeomorphism if ψ is an isometric isomorphism.

Secretly, what we have done here is defined two functors and two natural transformations. To see this, let $\mathcal{C}$

be the category with objects commutative C^* algebras and morphisms $*$-homomorphisms. Further, let $\mathcal{D}$ be the category with objects compact Hausdorff topological spaces and morphisms continuous functions. Then we have a functor $F : \mathcal{C} \to \mathcal{D}$ defined on objects by $F(A) = \Omega(A)$ for any C^*-algebra A and defined on morphisms $\psi : A \to B$ by $F(\psi) = \psi^* : \Omega(B) \to \Omega(A)$ and a functor $G : \mathcal{D} \to \mathcal{C}$ defined on objects by $F(X) = C(X)$ for any compact Hausdorff space X and defined on morphisms $\psi : X \to Y$ by $G(\psi) = \psi^* : C(Y) \to C(X)$. The fact that these are functors is the statment of Proposition 7.2. Also notice that these functors are contravariant. Additionally, the Gelfand transform $\mathcal{G}$ and the map $\mathcal{F}$ defined in Theorem 7.1 are secretely natural isomorphisms $\mathcal{G} : \mathrm{id}_{\mathcal{C}} \to GF$ and $\mathcal{F} : \mathrm{id}_{\mathcal{D}} \to FG$.

Theorem 7.3: The Gelfand transform $\mathcal{G}$ and its inverse $\mathcal{F}$ are natural isomorphisms $\mathcal{G} : \mathrm{id}_{\mathcal{C}} \to GF$ and $\mathcal{F} : \mathrm{id}_{\mathcal{D}} \to FG$ and thus $\mathcal{C}$ and $\mathcal{D}$ are equivalent categories.

Proof. From Theorem 6.7, we have that $\mathcal{G}$ and $\mathcal{F}$ are component-wise isomorphisms, so we only need to show that $\mathcal{G}$ and $\mathcal{F}$ are natural. Let $\psi : A \to B$ be a morphism between C^*-algebras.

$$\begin{array}{ccc} A & \xrightarrow{\mathcal{G}_A} & C(\Omega(A)) \\ {\scriptstyle\psi}\downarrow & & \downarrow{\scriptstyle GF(\psi)} \\ B & \xrightarrow{\mathcal{G}_B} & C(\Omega(B)) \end{array}$$

To show that $\mathcal{G}$ is natural, we need to show that this diagram commutes. We do this by unravelling definitions. Let $\omega \in \Omega(B)$.

$$\begin{aligned} \left[GF(\psi)(\mathcal{G}_A(a))\right](\omega) &= \left[GF(\psi)(\hat{a})\right](\omega) = \left[\hat{a} \circ F(\psi)\right](\omega) \\ &= \hat{a}(\omega \circ \psi) = (\omega \circ \psi)(a) \\ &= \widehat{\psi(a)}(\omega) = \mathcal{G}_B(\psi(a))(\omega) \end{aligned}$$

Thus $\mathcal{G}$ is a natural isomorphism. Finally, let $\phi : X \to Y$ be a morphism of compact topological spaces.

$$\begin{array}{ccc} X & \xrightarrow{\mathcal{F}_X} & \Omega(C(X)) \\ {\scriptstyle\phi}\downarrow & & \downarrow{\scriptstyle FG(\phi)} \\ Y & \xrightarrow{\mathcal{F}_Y} & \Omega(C(Y)) \end{array}$$

We show that $\mathcal{F}$ is natural by showing that this diagram commutes. Let $g \in C(Y)$.

$$\begin{aligned} [FG(\phi)(\mathcal{F}_X(x))](g) &= [FG(\phi)(\hat{x})](g) = [\hat{x} \circ G(\phi)](g) \\ &= \hat{x}(g \circ \phi) = g(\phi(x)) \\ &= \widehat{\phi(x)}(g) = \mathcal{F}_Y(\phi(x))(g) \end{aligned}$$

Therefore, the diagram commutes and $\mathcal{F}$ is a natural isomorphism. That is, $\mathcal{G}$ and $\mathcal{F}$ form an equivalence of categories between $\mathcal{C}$ and $\mathcal{D}$. □

In summary, we have proven more than just the statement of the Gelfand Duality (along with all of its intricate details). We have constructed the inverse transformation, and shown that these are natural isomorphisms that form an equivalence of categories between commutative C^*-algebras and compact, Hausdorff topological spaces. That is, we can essentially think of commutative C^*-algebras as being the same as compact, Hausdorff spaces, and we have shown this formally in a categorical sense.

References

[1] Vahid Shirbisheh. (14 November 2012). Lectures on C^*-algebras. Available: https://arxiv.org/pdf/1211.3404.pdf.

All propositions, proofs, lemmas, theorems and the majority of what I learnt (except the proofs of Proposition 3.1, Lemma 4.2 and Proposition 4.5, which I came up with myself) came from the source above. However the work I have submitted is my own and written from my own understanding.

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[4] Alexander Isaev. (2017). Complex Analysis Lecture Notes for MATH3228 and MATH6213.

All of the complex analysis mentioned in this essay was covered in MATH3228 - Advanced Complex Analysis (at least in the 2017 version of the course). Hence Alex's notes are a good reference for this section (I have a copy, but I am not sure if they are publicly available online anywhere).

[5] Stefano Ferri. Convergence in Topological Spaces. Nets. Available: https://matematicas.uniandes.edu.co/~stferri/netsfunctional.pdf.

Some proofs required some reference to point-set topology that we have not formally learnt, however they are covered in this above source. We have only used very basic notions such as convergent nets and using them to prove sets are closed or functions are continuous.

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